

Examination Regulations

Only the following may be used during the examination:

- Microsoft SQL Server.
- SQL Server Management Studio (SSMS) without any AI features, AI assistants, AI plug-ins, or AI extensions enabled or installed.
- Printed materials or offline documents that are already stored on your computer before the examination.
- A network connection **only** to the **ExamOnline** Wi-Fi network for examination purposes.

The following are strictly prohibited:

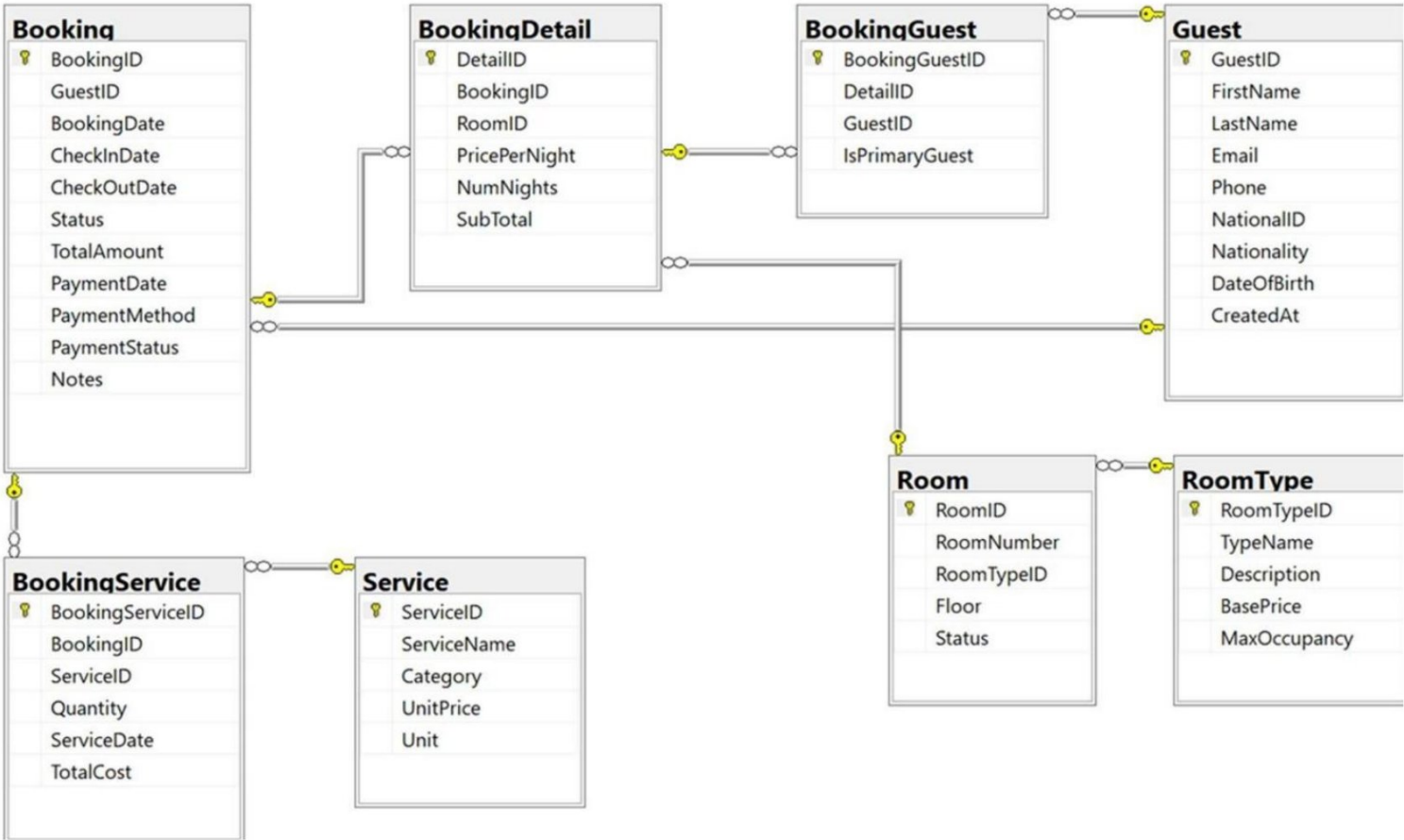
- Connecting to any network other than the **ExamOnline** Wi-Fi network, including personal hotspots, mobile data, VPNs, remote desktop services, cloud storage, or any other Internet connection.
- Accessing any website or online service that is not required for the examination.
- Any online or offline AI tools, including but not limited to ChatGPT, Microsoft Copilot, GitHub Copilot, Gemini, Claude, DeepSeek, Cursor AI, Codeium, Tabnine, Amazon Q, NotebookLM, LMStudio, Visual Studio Code (VSCode), or any other AI coding assistant.
- Any AI-powered extensions, plug-ins, browser add-ons, or IDE/SSMS extensions that provide code generation, code completion, explanations, or suggestions.
- Any communication or file sharing with other candidates or external parties.

Violation Policy

Failing to comply with any of the requirements or regulations above is considered academic misconduct and will result in an examination score of 0.

Database used from Question 2 onward

From Question 2 onward, use the database provided in the accompanying .sql file. Before answering the questions, run the provided script to create the required tables and populate them with the given data. The database schema is shown in the following diagram.



(Hình ảnh: Sơ đồ cơ sở dữ liệu dạng SSMS Database Diagram gồm 8 bảng với các cột và khóa chính (ký hiệu chìa khóa) cùng các đường quan hệ (ký hiệu one/many). Các bảng và thuộc tính:

- **Booking** (PK BookingID): BookingID, GuestID, BookingDate, CheckInDate, CheckOutDate, Status, TotalAmount, PaymentDate, PaymentMethod, PaymentStatus, Notes.
- **BookingDetail** (PK DetailID): DetailID, BookingID, RoomID, PricePerNight, NumNights, SubTotal.
- **BookingGuest** (PK BookingGuestID): BookingGuestID, DetailID, GuestID, IsPrimaryGuest.
- **Guest** (PK GuestID): GuestID, FirstName, LastName, Email, Phone, NationalID, Nationality, DateOfBirth, CreatedAt.
- **Room** (PK RoomID): RoomID, RoomNumber, RoomTypeID, Floor, Status.
- **RoomType** (PK RoomTypeID): RoomTypeID, TypeName, Description, BasePrice, MaxOccupancy.
- **BookingService** (PK BookingServiceID): BookingServiceID, BookingID, ServiceID, Quantity, ServiceDate, TotalCost.
- **Service** (PK ServiceID): ServiceID, ServiceName, Category, UnitPrice, Unit.

Các đường quan hệ (khóa ngoại -> khóa chính): Booking.GuestID -> Guest.GuestID; Booking.BookingID -> BookingDetail.BookingID; Booking.BookingID ->

Question 1

Create a database, then write SQL statements to create, inside this database, all the tables derived from the ERD shown in the picture below, complete with appropriate attributes, primary keys, and foreign keys.

NOTE that when writing the SQL commands, you MUST keep the names of the tables, relationships, attributes, and data types of attributes exactly as given in the ERD.

Attributes with an underline are the primary key of each entity.

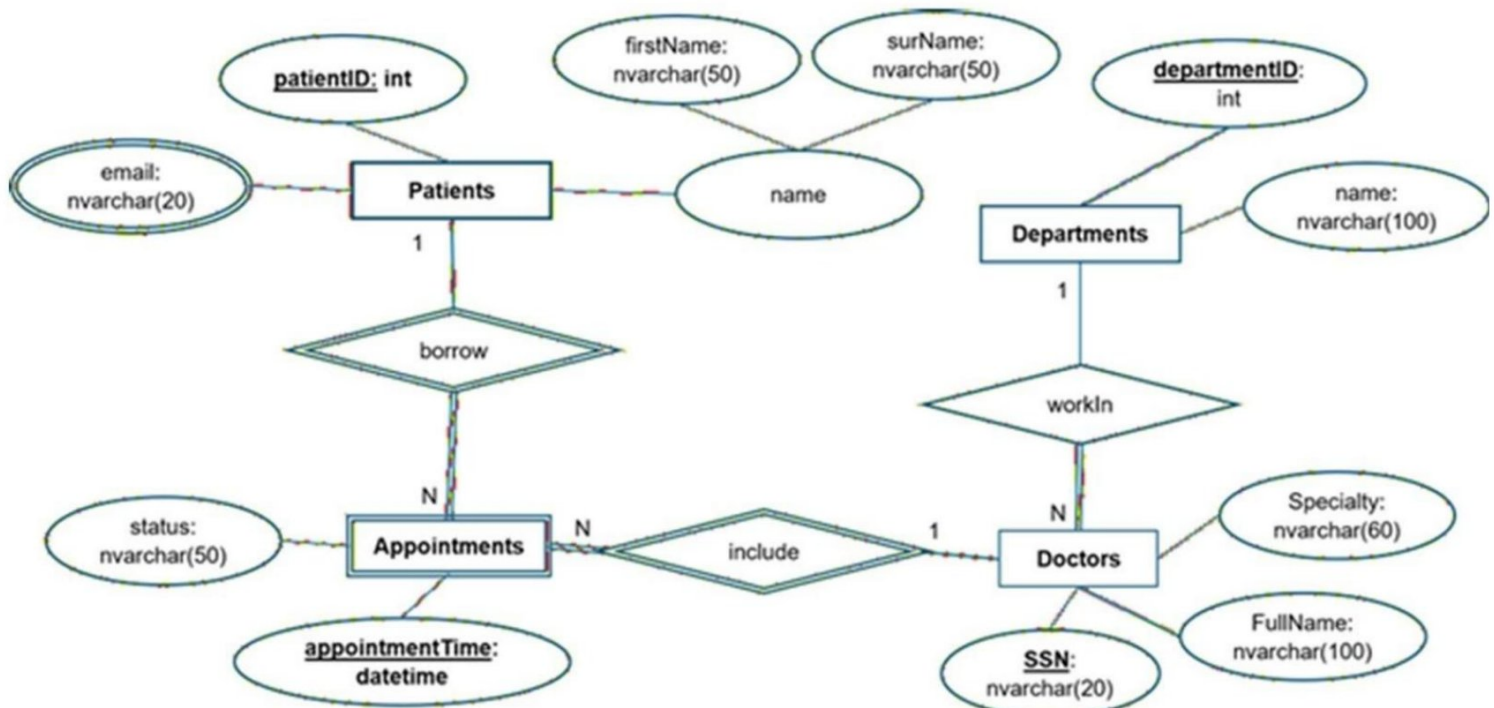
Attributes that reference the primary key of another table must have the same name as the attributes in that primary key.

If a table is created to represent a relationship, its name must be the same as the relationship's name.

When answering this question, submit only the SQL statements that create the tables with their corresponding keys and foreign keys. Do not use "create database", "Alter database", "use database_name", or any statement that uses the database's name in your submission.

If subclasses exist in this diagram, you must apply ER style conversions to them.

A table that represents a multi-valued attribute must be named by concatenating the table name and the attribute name. For example, if a table named Student has a multi-valued attribute named Phone, the corresponding table must be named StudentPhone.



[Hình ảnh: Sơ đồ ERD (Picture 1.1) gồm 4 thực thể và 3 quan hệ.

Thực thể **Patients** (hình chữ nhật viền đơn): thuộc tính patientID: int (gạch chân = PK); thuộc tính email: nvarchar(20) (hình elip viền đôi = thuộc tính đa trị); thuộc tính ghép name gồm firstName: nvarchar(50) và surName: nvarchar(50).

Thực thể **Departments** (viền đơn): thuộc tính departmentID: int (gạch chân = PK); thuộc tính name: nvarchar(100).

Thực thể **Appointments** (hình chữ nhật viền đôi): thuộc tính status: nvarchar(50); thuộc tính appointmentTime: datetime (gạch chân = PK).

Thực thể **Doctors** (viền đơn): thuộc tính SSN: nvarchar(20) (gạch chân = PK); thuộc tính Specialty: nvarchar(60); thuộc tính FullName: nvarchar(100).

Quan hệ **borrow** (hình thoi viền đôi) nối Patients (lực lượng 1) với Appointments (lực lượng N).

Quan hệ **workIn** (hình thoi viền đơn) nối Departments (lực lượng 1) với Doctors (lực lượng N).

Quan hệ **include** (hình thoi viền đôi) nối Appointments (lực lượng N) với Doctors (lực lượng 1).]

Question 2

Write an SQL query to list all services whose unit price is between 10 and 30 (inclusive), as shown below:

#	ServiceID	ServiceName	Category	UnitPrice	Unit
1	1	Breakfast Buffet	Food & Beverage	15.00	per person per day
2	2	Breakfast Room Service	Food & Beverage	10.00	per order
3	3	Airport Transfer	Transport	25.00	per trip
4	6	Mini Bar Refill	Food & Beverage	20.00	per refill

5	10	Business Center	Business	10.00	per hour
---	----	-----------------	----------	-------	----------

Picture 2.1

Question 3

Write an SQL query to show the BookingID, GuestName (FirstName concatenated with a single space and LastName), Nationality, BookingDate, DetailID, and RoomNumber for all bookings whose BookingDate is in 2024 and whose guests' nationality is Chinese or Japanese, as shown below:

#	BookingID	GuestName	Nationality	BookingDate	DetailID	RoomNumber
1	9	Chen Wei	Chinese	2024-09-16 00:00:00.000	18	604
2	10	Chen Wei	Chinese	2024-03-24 00:00:00.000	19	605
3	10	Chen Wei	Chinese	2024-03-24 00:00:00.000	20	606
4	11	Chen Wei	Chinese	2024-10-26 00:00:00.000	21	701
5	11	Chen Wei	Chinese	2024-10-26 00:00:00.000	22	702
6	11	Chen Wei	Chinese	2024-10-26 00:00:00.000	23	703
7	12	Chen Wei	Chinese	2024-05-28 00:00:00.000	24	704
8	17	Hiroshi Tanaka	Japanese	2024-05-21 00:00:00.000	33	201
9	17	Hiroshi Tanaka	Japanese	2024-05-21 00:00:00.000	34	202
10	17	Hiroshi Tanaka	Japanese	2024-05-21 00:00:00.000	35	301
11	19	Hiroshi Tanaka	Japanese	2024-07-31 00:00:00.000	37	401
12	19	Hiroshi Tanaka	Japanese	2024-07-31 00:00:00.000	38	402
13	20	Hiroshi Tanaka	Japanese	2024-03-02 00:00:00.000	39	403
14	20	Hiroshi Tanaka	Japanese	2024-03-02 00:00:00.000	40	404
15	20	Hiroshi Tanaka	Japanese	2024-03-02 00:00:00.000	41	501
16	33	Linda Le	Chinese	2024-10-17 00:00:00.000	66	302
17	34	Linda Le	Chinese	2024-05-19 00:00:00.000	67	401
18	34	Linda Le	Chinese	2024-05-19 00:00:00.000	68	402
19	39	Yuki Suzuki	Japanese	2024-05-04 00:00:00.000	78	604
20	40	Yuki Suzuki	Japanese	2024-12-06 00:00:00.000	79	605
21	40	Yuki Suzuki	Japanese	2024-12-06 00:00:00.000	80	606
22	41	Yuki Suzuki	Japanese	2024-07-08 00:00:00.000	81	701
23	41	Yuki Suzuki	Japanese	2024-07-08 00:00:00.000	82	702
24	41	Yuki Suzuki	Japanese	2024-07-08 00:00:00.000	83	703

— Tài liệu được thành viên đăng tải tại **FUOVERFLOW** và được biên soạn lại để hỗ trợ ôn tập —

Picture 3.1

Question 4

Write an SQL query to show RoomTypeID, TypeName, Description, RoomNumber, Floor, DetailID, BookingGuestID, FirstName, LastName, and Nationality for every room type whose TypeName is 'Standard' or 'Suite'.

For each such room type, display its rooms, the booking details for those rooms, and the Japanese guests assigned to each booking detail. The results must include:

- every Standard and Suite room type, even if it has no rooms;
- every room of those types, even if it has no booking details;

In all such cases, the missing columns should appear as NULL.

Sort the output by TypeName in descending order, then by DetailID in descending order, and finally by BookingGuestID in ascending order, as shown below:

#	RoomTypeID	TypeName	Description	RoomNumber	Floor	DetailID	BookingGuestID	FirstName	LastName	Nationality
---	------------	----------	-------------	------------	-------	----------	----------------	-----------	----------	-------------

1	3	Suite	Luxury suite with living area and minibar	602	6	166	351	Kenji	Yamamoto	Japanese
2	3	Suite	Luxury suite with living area and minibar	602	6	166	352	Sakura	Nakamura	Japanese
3	3	Suite	Luxury suite with living area and minibar	606	6	140	298	Yuki	Suzuki	Japanese
4	3	Suite	Luxury suite with living area and minibar	601	6	135	285	Sakura	Nakamura	Japanese
5	3	Suite	Luxury suite with living area and minibar	605	6	109	229	Hiroshi	Tanaka	Japanese
6	3	Suite	Luxury suite with living area and minibar	603	6	47	97	Hiroshi	Tanaka	Japanese
7	3	Suite	Luxury suite with living area and minibar	606	6	20	40	Kenji	Yamamoto	Japanese
8	3	Suite	Luxury suite with living area and minibar	602	6	16	30	Hiroshi	Tanaka	Japanese
9	3	Suite	Luxury suite with living area and minibar	604	6	NULL	NULL	NULL	NULL	NULL
10	1	Standard	Cozy room with basic amenities	101	1	181	389	Kenji	Yamamoto	Japanese
11	1	Standard	Cozy room with basic amenities	301	3	155	331	Hiroshi	Tanaka	Japanese
12	1	Standard	Cozy room with basic amenities	301	3	125	267	Yuki	Suzuki	Japanese
13	1	Standard	Cozy room with basic amenities	101	1	121	259	Yuki	Suzuki	Japanese
14	1	Standard	Cozy room with basic amenities	201	2	63	134	Sakura	Nakamura	Japanese
15	1	Standard	Cozy room with basic amenities	102	1	62	133	Hiroshi	Tanaka	Japanese
16	1	Standard	Cozy room with basic amenities	101	1	61	130	Kenji	Yamamoto	Japanese
17	1	Standard	Cozy room with basic amenities	301	3	5	9	Kenji	Yamamoto	Japanese
18	1	Standard	Cozy room with basic amenities	101	1	1	2	Yuki	Suzuki	Japanese
19	1	Standard	Cozy room with basic amenities	202	2	NULL	NULL	NULL	NULL	NULL
20	1	Standard	Cozy room with basic amenities	302	3	NULL	NULL	NULL	NULL	NULL

— Tài liệu được thành viên đăng tải tại **FUOVERFLOW** và được biên soạn lại để hỗ trợ ôn tập —

Picture 4.1

Question 5

Write a SQL query that generates a report on the performance of each room type for January 2024. The report must include the following columns:

- RoomTypeID - the room type identifier.
- TypeName - the room type name.
- TotalBookings - the total number of distinct bookings that checked in during January 2024.
- TotalRevenue - the total revenue generated from bookings that checked in during January 2024.
- AveragePricePerNight - the average revenue earned per occupied night, calculated as:

TotalRevenue/TotalNumberOfNights, where the total number of nights (TotalNumberOfNights) is the sum of NumNights for all bookings that checked in during January 2024 of the room type.

Requirements:

- Consider only bookings with a CheckInDate in January 2024.
- Include all room types, even if they have no bookings during this period.
- Display 0 for TotalRevenue and AveragePricePerNight when there are no bookings.
- Display AveragePricePerNight as a DECIMAL(10,2) value.
- Sort the result by TotalRevenue in descending order. If two room types have the same revenue, sort them by TypeName in ascending order.

#	RoomTypeID	TypeName	TotalBookings	TotalRevenue	AveragePricePerNight
1	2	Deluxe	2	4410.00	163.33
2	4	Family	1	2520.00	210.00
3	1	Standard	3	1808.00	86.10
4	5	Penthouse	1	1200.00	600.00
5	3	Suite	0	0.00	0.00

Picture 5.1

Question 6

Write a SQL query that generates a report summarizing booking activity by guest nationality. For each nationality, show the following information:

- Nationality - the nationality of the guests.
- NumberOfGuests - the total number of guests with that nationality (include all guests, even those who have never made a booking).
- MinBookingsPerGuest - the minimum number of bookings made by any guest of that nationality.
- MaxBookingsPerGuest - the maximum number of bookings made by any guest of that nationality.

#	Nationality	NumberOfGuests	MinBookingsPerGuest	MaxBookingsPerGuest
1	American	4	1	4
2	Australian	2	1	1
3	British	5	1	4
4	Chinese	4	1	4
5	Egyptian	2	3	3
6	French	2	3	3
7	German	2	1	1
8	Ghanaian	1	0	0
9	Hungarian	1	2	2
10	Indian	3	0	2
11	Italian	2	1	1
12	Japanese	5	0	4
13	Jordanian	2	1	2
14	Korean	2	0	0
15	Lebanese	1	2	2
16	Norwegian	1	2	2
17	Omani	1	1	1
18	Russian	2	1	1
19	Spanish	3	3	4
20	Swedish	1	2	2
21	Vietnam...	5	2	4

Picture 6.1

Question 7

Write an SQL query to identify the guest(s) with the highest total revenue for each nationality in 2024. For each nationality, show the guest(s) whose total revenue (computed as the sum of room charges (SubTotal) and service charges (TotalCost)) from all of his/her bookings with a CheckInDate in 2024 is the highest among guests of the same nationality. Guests who had no bookings in 2024 must still be included, with their TotalRevenue reported as 0.

For each result row, display:

- Nationality
- GuestID
- GuestName (the concatenation of the guest's first name, a single space, and the last name)
- TotalRevenue (room charges + service charges)

If several guests of the same nationality share the same highest TotalRevenue, include all of them in the result.

Sort the output by Nationality in ascending order and then by GuestID in ascending order for guests of the same nationality.

#	Nationality	GuestID	GuestName	TotalRevenue
1	American	2	Bob Smith	6650.00
2	Australian	35	Luke Anderson	0.00
3	Australian	43	Alex Taylor	0.00
4	British	7	James Wilson	10040.00
5	Chinese	3	Chen Wei	6289.00
6	Egyptian	16	Fatima Hassan	9968.00
7	French	17	Pierre Dupont	7190.00
8	German	41	Thomas Mueller	0.00
9	German	42	Claire Wagner	0.00
10	Ghanaian	50	Aisha Osei	0.00
11	Hungarian	22	Anna Kovacs	0.00
12	Indian	23	Ravi Sharma	0.00
13	Indian	24	Priya Patel	0.00
14	Indian	49	Raj Singh	0.00
15	Italian	29	Marco Rossi	0.00
16	Italian	30	Giulia Ferrari	0.00
17	Japanese	11	Yuki Suzuki	8448.00
18	Jordanian	27	Omar Ibrahim	0.00
19	Jordanian	34	Amira Khalil	0.00
20	Korean	47	Lee Park	0.00
21	Korean	48	Kim Choi	0.00
22	Lebanese	28	Layla Mansour	0.00
23	Norwegian	26	Astrid Hansen	0.00
24	Omani	33	Hassan Al-Far...	0.00
25	Russian	37	Ivan Petrov	0.00
26	Russian	38	Natasha Ivano...	0.00
27	Spanish	4	Maria Garcia	17428.00
28	Swedish	25	Lars Eriksson	0.00
29	Vietnam...	8	Sophie Pham	4899.00

Picture 7.1

Question 8

Write a stored procedure called `addService` that adds a service to an existing booking.

— Tài liệu được thành viên đăng tải tại **FUOVERFLOW** và được biên soạn lại để hỗ trợ ôn tập —

The procedure has the following input parameters:

- `@bookingID INT`: the first parameter, the booking ID.
- `@serviceID INT`: the second parameter, the service ID.
- `@quantity INT`: the third parameter, the quantity of the service.

The procedure must satisfy the following requirements:

- Validate the input data:
 - The `bookingID` must exist in the `Booking` table.
 - The `serviceID` must exist in the `Service` table.
 - The quantity must be greater than or equal to 0.
 - If any validation fails:
 - \$ Print the message "Input values are not valid."
 - \$ Do not insert any record into the `BookingService` table.
 - \$ And return 1
 - Otherwise:
 - \$ Insert a new record into the `BookingService` table with the corresponding parameters `@bookingID`, `@serviceID`, `@quantity`. The `TotalCost` must be calculated as `Quantity x UnitPrice` of the corresponding service.
 - \$ And return 0

For example, when the `addService` procedure is executed with the following statements, it returns 1 in the first three scenarios because the input parameters are not valid. In the last scenario, it returns 0 and a new row is inserted into the `BookingService` table with `bookingID = 100`, `serviceID = 10`, `quantity = 2`, `TotalCost = 20`, and `ServiceDate` set to its default value (`getdate()`).

```
-- Scenario 1: status = 1, no row is inserted
declare @status int
exec @status = addService 1000, 5, 2
select @status as Status

-- Scenario 2: status = 1, no row is inserted
declare @status int
exec @status = addService 99, 15, 2
select @status as Status

-- Scenario 3: status = 1, no row is inserted
declare @status int
exec @status = addService 99, 5, -2
select @status as Status

-- Scenario 4: status = 0 and new row is inserted
declare @status int
exec @status = addService 100, 10, 2
select @status as Status
select *
from BookingService
```

Question 9

1 Write a SQL Server trigger named `trg_PreventRoomOverbooking` on the `BookingDetail` table. The trigger runs when an insert statement inserts one or more rows into the `BookingDetail` table. For every inserted row, the trigger must satisfy the following requirements:

- Automatically calculate `NumNights` as the number of days between the corresponding booking's `CheckInDate` and `CheckOutDate`. Ignore the value supplied by the user.
- Prevent inserting a booking detail if:
 - The `BookingID` does not exist in the `Booking` table.
 - The `RoomID` does not exist in the `Room` table.
 - The same `RoomID` already exists in the same booking.
 - The room has already been assigned to another booking whose `Status` is not 'Cancelled' and whose booking period (from `CheckInDate` to `CheckOutDate`) overlaps with the new booking's stay.
 - If an inserted row violates either rule above, it must not be inserted.
 - Rows that satisfy all requirements should be inserted successfully. Valid rows should still be inserted, even if other rows in the same `INSERT` statement fail the trigger conditions.

After creating the trigger, run the following test statement; the result should be as follows:

— Tài liệu được thành viên đăng tải tại **FUOVERFLOW** và được biên soạn lại để hỗ trợ ôn tập —

- The first row must be ignored because the `bookingID = 150` does not exist in the `Booking` table and the `roomID = 50` does not exist in the `Room` table.

- The second row must be ignored because the room with roomID = 1 already exists in the booking with bookingID = 1.
- The third row must be ignored because the room with roomID = 3 already exists in another booking whose booking period overlaps with the booking with bookingID = 19.
- The final row will be inserted but with the NumNights = 2 as in the following figure because the booking with bookingID = 2 stays only 2 nights (not 4 nights).

```
INSERT BookingDetail(BookingID,RoomID, PricePerNight, NumNights)
VALUES (150,50,50,1), -- invalid bookingID and RoomID
(1,1,88,1), -- room already in the booking
(19,3,80, 4), -- overlap checkInDate and CheckOutDate with another booking
(2,1,88,4) -- valid insert data

select * from BookingDetail
```

#	DetailID	BookingID	RoomID	PricePerNight	NumNights	SubTotal
197	197	98	17	360.00	7	2520.00
198	198	99	18	300.00	1	300.00
199	199	100	19	300.00	2	600.00
200	200	100	20	330.00	2	660.00
201	201	2	1	88.00	2	176.00

Picture 9.1

Question 10

Write a SQL statement to update the TotalAmount column of the bookings whose BookingID is from 1 to 10 (inclusive).

The TotalAmount of each booking is calculated as:

- the total room revenue (sum of SubTotal in the BookingDetail table), plus
- the total service charges (sum of TotalCost in the BookingService table).

When a booking has no room details or no services, the corresponding total is treated as 0.

After the update, the first 20 rows of the Booking table are as follows.

#	BookingID	GuestID	BookingDate	CheckInDate	CheckOutDate	Status	TotalAmount	PaymentDate	PaymentMethod	PaymentStatus
1	1	1	2023-12-31 00:00:00.000	2024-01-01	2024-01-02	Checked-Out	176.00	2024-01-02 00:00:00.000	Credit Card	Completed
2	2	1	2024-08-02 00:00:00.000	2024-08-04	2024-08-06	Checked-Out	549.00	2024-08-06 00:00:00.000	Debit Card	Completed
3	3	1	2024-03-04 00:00:00.000	2024-03-07	2024-03-10	Checked-Out	284.00	2024-03-10 00:00:00.000	Bank Transfer	Completed
4	4	1	2024-10-06 00:00:00.000	2024-10-10	2024-10-14	Confirmed	1328.00	2024-10-14 00:00:00.000	Online	Pending
5	5	2	2024-05-08 00:00:00.000	2024-05-13	2024-05-18	Confirmed	2475.00	2024-05-18 00:00:00.000	Cash	Pending
6	6	2	2024-12-10 00:00:00.000	2024-12-16	2024-12-22	Checked-In	900.00	2024-12-22 00:00:00.000	Credit Card	Completed
7	7	2	2024-07-12 00:00:00.000	2024-07-19	2024-07-26	Cancelled	2215.00	2024-07-26 00:00:00.000	Debit Card	Refunded
8	8	2	2024-02-14 00:00:00.000	2024-02-22	2024-02-23	No-Show	1060.00	2024-02-23 00:00:00.000	Bank Transfer	Failed
9	9	3	2024-09-16 00:00:00.000	2024-09-25	2024-09-27	Confirmed	730.00	2024-09-27 00:00:00.000	Online	Pending
10	10	3	2024-03-24 00:00:00.000	2024-04-03	2024-04-06	Checked-Out	1890.00	2024-04-06 00:00:00.000	Cash	Completed
11	11	3	2024-10-26 00:00:00.000	2024-11-06	2024-11-10	Checked-Out	0.00	2024-11-10 00:00:00.000	Credit Card	Completed
12	12	3	2024-05-28 00:00:00.000	2024-06-09	2024-06-14	Checked-Out	0.00	2024-06-14 00:00:00.000	Debit Card	Completed

13	13	4	2023-12-30 00:00:00.000	2024-01-12	2024-01-18	Checked- Out	0.00	2024-01-18 00:00:00.000	Bank Transfer	Completed
14	14	4	2024-08-01 00:00:00.000	2024-08-15	2024-08-22	Confirmed	0.00	2024-08-22 00:00:00.000	Online	Pending
15	15	4	2024-03-17 00:00:00.000	2024-03-18	2024-03-19	Confirmed	0.00	2024-03-19 00:00:00.000	Cash	Pending
16	16	4	2024-10-19 00:00:00.000	2024-10-21	2024-10-23	Checked- In	0.00	2024-10-23 00:00:00.000	Credit Card	Completed
17	17	5	2024-05-21 00:00:00.000	2024-05-24	2024-05-27	Cancelled	0.00	2024-05-27 00:00:00.000	Debit Card	Refunded
18	18	5	2023-12-29 00:00:00.000	2024-01-02	2024-01-06	No-Show	0.00	2024-01-06 00:00:00.000	Bank Transfer	Failed
19	19	5	2024-07-31 00:00:00.000	2024-08-05	2024-08-10	Confirmed	0.00	2024-08-10 00:00:00.000	Online	Pending
20	20	5	2024-03-02 00:00:00.000	2024-03-08	2024-03-14	Checked- Out	0.00	2024-03-14 00:00:00.000	Cash	Completed

— Tài liệu được thành viên đăng tải tại **FUOVERFLOW** và được biên soạn lại để hỗ trợ ôn tập —

Picture 10.1